

Bayesian Logistic Regression

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1 Classification Problems

In the regression problems that we have studied so far, the response variable was quantitative. For example, we might try to predict a person’s blood pressure, a student’s score on an exam, or the price of a house. In a classification problem, the response variable is qualitative. It takes values in a finite set of classes. For example,

$$Y \in \{\text{spam}, \text{not spam}\},$$

or, in a multiclass problem,

$$Y \in \{\text{setosa}, \text{versicolor}, \text{virginica}\}.$$

The goal is to use a feature vector X to predict the class of Y . Thus, a classifier is a function

$$C(x) \in \mathcal{C},$$

where $\mathcal{C}$ is the set of possible classes. In many statistical and machine learning applications, however, we do not only want the final predicted class. We often want the probabilities of the different classes. For example, it is usually more useful to know that the probability of default is 0.42 than to be told only “default” or “no default.”

Thus, in a binary classification problem, the basic object we want to estimate is

$$P(Y = 1 \mid X = x).$$

Once this probability has been estimated, we can turn it into a classification rule by choosing a threshold.

1.1 Example: Credit Card Default

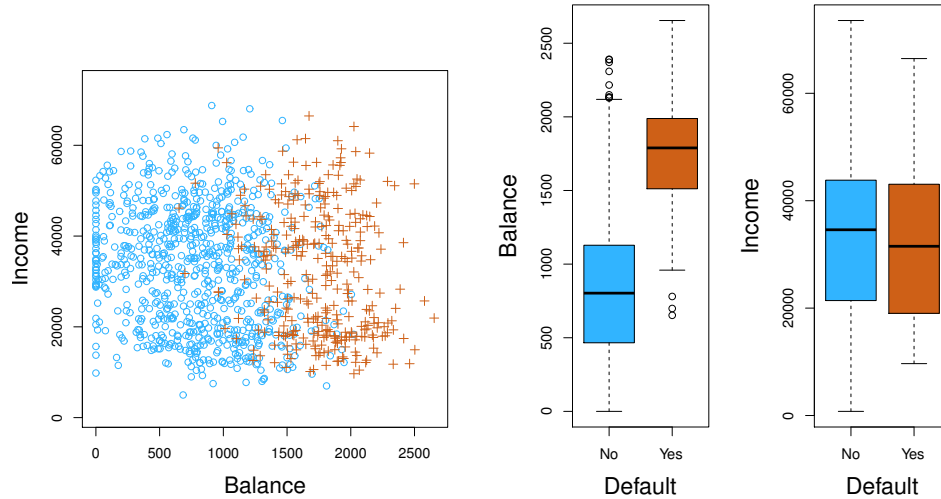
Let us see a credit card default data set. The response variable is whether a customer defaulted on their credit card debt. The predictors include variables such as balance, income, and student status.

Figure 1 suggests a natural statistical question. Given a customer’s balance and income, can we estimate the probability that the customer will default? In other words, can we estimate

$$P(\text{default} \mid \text{balance}, \text{income})?$$

This is a probability estimation problem. A final yes/no classification can be made afterwards, but the more fundamental object is the conditional probability of default.

Example: Credit Card Default



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Figure 1: Credit card default data. The plot and boxplots suggest that balance is strongly associated with default, while income is less clearly separated across the two groups.

1.2 Can We Use Linear Regression?

A first thought is to code the response numerically. For the default data, define

$$Y = \begin{cases} 1 & \text{if the customer defaults,} \\ 0 & \text{if the customer does not default.} \end{cases}$$

Now suppose we fit a linear regression of Y on the predictors. This idea is not as unreasonable as it may first seem. Indeed, because Y only takes the values 0 and 1, we have

$$E(Y \mid X = x) = P(Y = 1 \mid X = x).$$

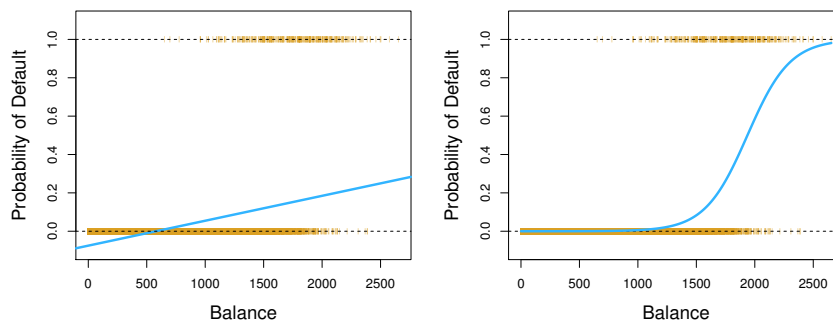
Thus, if we could model the conditional expectation of Y well, we would also be modeling the conditional probability of the positive class.

However, there is a problem. A linear regression model has the form

$$E(Y \mid X = x) = \beta_0 + \beta_1 x_1 + \cdots + \beta_m x_m.$$

The right hand side can be negative or larger than one. Therefore, linear regression can produce fitted values that cannot be interpreted as probabilities.

Linear versus Logistic Regression



The orange marks indicate the response Y , either 0 or 1. Linear regression does not estimate $\Pr(Y = 1|X)$ well. Logistic regression seems well suited to the task.

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Figure 2: Linear regression versus logistic regression for predicting probability of default from balance. The logistic curve stays between zero and one.

The issue is not that linear regression is always a terrible classifier in binary problems. In fact, it can sometimes classify reasonably well. The issue is that it is not an ideal model for probabilities. A probability should always lie between zero and one. Logistic regression fixes this problem by replacing the linear model for the probability with a linear model for the log-odds.

1.3 The Logistic Model

The key idea behind logistic regression is to start with a linear predictor

$$\eta(x) = \beta_0 + \beta_1 x_1 + \cdots + \beta_m x_m,$$

and then transform this linear predictor into a probability.

The transformation used in logistic regression is the logistic function

$$g(z) = \frac{e^z}{1 + e^z}.$$

Remark 1.1. *This seems somewhat arbitrary. We could have presumably used any other strictly increasing function bijectively mapping the real line to $(0, 1)$.*

Thus the logistic regression model is

$$P(Y = 1 \mid X = x) = \frac{\exp(\beta_0 + \beta_1 x_1 + \cdots + \beta_m x_m)}{1 + \exp(\beta_0 + \beta_1 x_1 + \cdots + \beta_m x_m)}.$$

Equivalently, if

$$p(x) = P(Y = 1 \mid X = x),$$

then

$$p(x) = \frac{e^{\eta(x)}}{1 + e^{\eta(x)}}.$$

The advantage of this model is immediate. No matter what value the linear predictor $\eta(x)$ takes, the fitted probability $p(x)$ always lies between zero and one.

For very negative values of the linear predictor, the probability is close to zero. For very positive values, the probability is close to one. In the middle, the curve changes most rapidly.

The logistic model has a particularly simple interpretation in terms of odds. The odds are

$$\frac{p(x)}{1 - p(x)}.$$

For example, if $p(x) = 0.8$, then the odds are

$$\frac{0.8}{0.2} = 4.$$

This means that the event is four times as likely to occur as not to occur.

For the logistic model, taking logarithms gives

$$\log \left(\frac{p(x)}{1 - p(x)} \right) = \beta_0 + \beta_1 x_1 + \cdots + \beta_m x_m.$$

Thus logistic regression is a linear model for the log-odds of the positive class.

This gives a useful interpretation of the coefficients. If x_j increases by one unit while all other variables remain fixed, then the log-odds increase by β_j . Equivalently, the odds are multiplied by e^{β_j} . The usual caveat of interpreting this as an adjusted coefficient (adjusting for the other covariates in the model) remains true here and in any generalized linear model (to be discussed later).

Remark 1.2. *We can fit a nonlinear function to log odds as well; using any nonlinear method we will learn later in the course.*

1.4 Fitting the Logistic Model

The logistic regression model contains an unknown parameter vector

$$\beta = (\beta_0, \beta_1, \dots, \beta_m)^T.$$

These parameters determine how the predictors influence the probability of belonging to the positive class. Once the parameters have been specified, the model assigns a probability

$$P(Y = 1 \mid X = x) = \frac{\exp(\beta_0 + \beta_1 x_1 + \dots + \beta_m x_m)}{1 + \exp(\beta_0 + \beta_1 x_1 + \dots + \beta_m x_m)}.$$

The unknown parameters are estimated from data using the method of *maximum likelihood* to obtain $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_m)^T$. We shall give a formal description of this procedure in the next section. The fitted model can then be used to estimate probabilities of default, spam, disease, or any other binary outcome of interest.

1.5 A Typical Software Output

After fitting a logistic regression model, statistical software typically produces output such as that shown below.

Maximum Likelihood

We use maximum likelihood to estimate the parameters.

$$\ell(\beta_0, \beta) = \prod_{i:y_i=1} p(x_i) \prod_{i:y_i=0} (1 - p(x_i)).$$

This *likelihood* gives the probability of the observed zeros and ones in the data. We pick β_0 and β_1 to maximize the likelihood of the observed data.

Most statistical packages can fit linear logistic regression models by maximum likelihood. In **R** we use the **glm** function.

	Coefficient	Std. Error	Z-statistic	P-value
Intercept	-10.6513	0.3612	-29.5	< 0.0001
balance	0.0055	0.0002	24.9	< 0.0001

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Figure 3: Logistic regression fit for the credit card default data.

Making Predictions

What is our estimated probability of **default** for someone with a balance of \$1000?

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X}} = \frac{e^{-10.6513 + 0.0055 \times 1000}}{1 + e^{-10.6513 + 0.0055 \times 1000}} = 0.006$$

With a balance of \$2000?

$$\hat{p}(X) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 X}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 X}} = \frac{e^{-10.6513 + 0.0055 \times 2000}}{1 + e^{-10.6513 + 0.0055 \times 2000}} = 0.586$$

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Figure 4: Estimated probabilities of default from the fitted logistic model.

Lets do it again, using **student** as the predictor.

	Coefficient	Std. Error	Z-statistic	P-value
Intercept	-3.5041	0.0707	-49.55	< 0.0001
student[Yes]	0.4049	0.1150	3.52	0.0004

$$\widehat{\Pr}(\text{default}=\text{Yes}|\text{student}=\text{Yes}) = \frac{e^{-3.5041+0.4049 \times 1}}{1 + e^{-3.5041+0.4049 \times 1}} = 0.0431,$$

$$\widehat{\Pr}(\text{default}=\text{Yes}|\text{student}=\text{No}) = \frac{e^{-3.5041+0.4049 \times 0}}{1 + e^{-3.5041+0.4049 \times 0}} = 0.0292.$$

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Figure 5: Estimated probabilities of default from the fitted logistic model.

Logistic regression with several variables

$$\log \left(\frac{p(X)}{1 - p(X)} \right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

$$p(X) = \frac{e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}$$

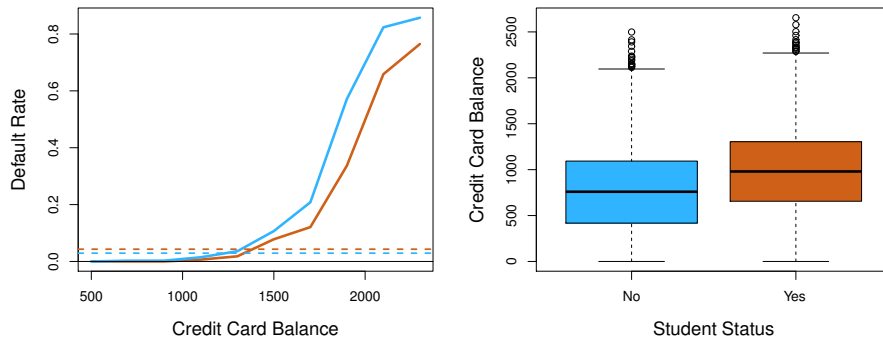
	Coefficient	Std. Error	Z-statistic	P-value
Intercept	-10.8690	0.4923	-22.08	< 0.0001
balance	0.0057	0.0002	24.74	< 0.0001
income	0.0030	0.0082	0.37	0.7115
student[Yes]	-0.6468	0.2362	-2.74	0.0062

Why is coefficient for **student** negative, while it was positive before?

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Figure 6: Estimated probabilities of default from the fitted logistic model.

Confounding



- Students tend to have higher balances than non-students, so their marginal default rate is higher than for non-students.
- But for each level of balance, students default less than non-students.
- Multiple logistic regression can tease this out.

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Figure 7: Estimated probabilities of default from the fitted logistic model.

Remark 1.3. *The above is an interesting example and shows that multiple regression works nicely to bring out the real effect of being a student on defaulting even though being a student is positively correlated with default rate. Again, in a multiple regression model, we should really interpret the coefficients as like univariate coefficients only after regressing out the other predictors in the model.*

The regression table contains quantities that should look familiar from linear regression. For each coefficient we see

- an estimate $\hat{\beta}_j$,
- a standard error,
- a z -statistic,
- and a p -value.

At first sight, these outputs raise several important questions. For example, where do the standard errors come from? Is the z -statistic approximately standard normal? The usual justification for the table is frequentist and is valid asymptotically (roughly when n is large). In the next section we develop the Bayesian inference underlying logistic

regression and show how the quantities reported by software can also be interpreted from this perspective.

1.6 Multiclass Logistic Regression

So far we have discussed binary classification. Logistic regression also has a multiclass version. Suppose we have K classes

$$Y \in \{1, 2, \dots, K\}.$$

We want to estimate all the conditional class probabilities

$$P(Y = 1 \mid X = x), \dots, P(Y = K \mid X = x),$$

which must be nonnegative and must sum to one.

One natural model is the softmax model:

$$P(Y = k \mid X = x) = \frac{\exp(\beta_{0k} + \beta_{1k}x_1 + \dots + \beta_{pk}x_p)}{\sum_{\ell=1}^K \exp(\beta_{0\ell} + \beta_{1\ell}x_1 + \dots + \beta_{p\ell}x_p)}.$$

This is also called multinomial logistic regression. We have a linear function for each class.

Remark 1.4. *There is some redundancy in the parameters here. We would really need $K - 1$ linear functions; however this form seems popular.*

The softmax form guarantees that all probabilities are positive and that they sum to one. The predicted class is often chosen to be the class with the largest estimated probability:

$$\hat{C}(x) = \operatorname{argmax}_{1 \leq k \leq K} \hat{P}(Y = k \mid X = x).$$

This same softmax transformation and multiclass logistic regression often appears in the last layer of modern neural networks. In a neural network classifier, the earlier layers produce features, and the last layer often uses a softmax to turn scores into class probabilities. In a language model, the same idea is used to assign probabilities to possible next tokens. We will see this later in the course.

2 Inference

We are again in the usual regression setting where we observe data

$$(y_i, x_{i1}, x_{i2}, \dots, x_{im}) \quad \text{for } i = 1, \dots, n.$$

There are m explanatory variables $x_1, \dots, x_m$ and one response variable. Suppose now that the response variable is binary, so that

$$y_i \in \{0, 1\}.$$

Let

$$x_i = (1, x_{i1}, \dots, x_{im})^T$$

and

$$\beta = (\beta_0, \beta_1, \dots, \beta_m)^T.$$

Then the model can be written compactly as

$$y_i \stackrel{\text{independent}}{\sim} \text{Bernoulli} \left(\frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)} \right).$$

2.1 Bayesian Inference

The likelihood is

$$L(\beta) = \prod_{i=1}^n \left(\frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)} \right)^{y_i} \left(1 - \frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)} \right)^{1-y_i}.$$

This can be simplified to

$$L(\beta) = \prod_{i=1}^n \frac{\exp(y_i x_i^T \beta)}{1 + \exp(x_i^T \beta)}.$$

The log-likelihood is therefore

$$\ell(\beta) := \sum_{i=1}^n [y_i(x_i^T \beta) - \log(1 + \exp(x_i^T \beta))].$$

Suppose $x_1^T, \dots, x_n^T$ form the rows of the $n \times p$ design matrix X , where $p = m + 1$. The unknown parameters in the logistic regression model are $\beta_0, \dots, \beta_m$. In contrast to the usual linear regression model, there is no additional variance parameter σ^2 .

As in linear regression, we may use the prior

$$\beta_0, \beta_1, \dots, \beta_m \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(-C, C) \quad (1)$$

for a large C . The posterior density of β is then

$$\begin{aligned} f_{\beta|\text{data}}(\beta) &\propto L(\beta) \times \text{prior} \\ &\propto \left[\prod_{i=1}^n \frac{\exp(y_i x_i^T \beta)}{1 + \exp(x_i^T \beta)} \right] I\{\beta_0, \beta_1, \dots, \beta_m \in (-C, C)\} \\ &= \exp(\ell(\beta)) I\{\beta_0, \beta_1, \dots, \beta_m \in (-C, C)\}. \end{aligned}$$

This posterior density is not a familiar standard density. If $p = 1$ or $p = 2$, one can plot it. In higher dimensions, one can use MCMC to sample from it. Here, however, we derive a multivariate normal approximation to the posterior. This is called the Laplace approximation or posterior normal approximation.

2.2 Normal Approximation to the Posterior

To get the normal approximation, first drop the indicator in the prior. For large C , this will typically be irrelevant near the high posterior probability region. Thus

$$f_{\beta|Y=y}(\beta) \propto \exp(\ell(\beta)).$$

The normal approximation is obtained by a second-order Taylor expansion of $\ell(\beta)$ around the MLE $\hat{\beta}$, defined by

$$\hat{\beta} := \operatorname{argmax}_{\beta \in \mathbb{R}^p} \ell(\beta).$$

The gradient of the log-likelihood is

$$\nabla \ell(\beta) = \sum_{i=1}^n \left(y_i - \frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)} \right) x_i. \quad (2)$$

The Hessian is

$$H\ell(\beta) = - \sum_{i=1}^n \frac{\exp(x_i^T \beta)}{(1 + \exp(x_i^T \beta))^2} x_i x_i^T.$$

Taylor expansion around $\hat{\beta}$ gives

$$\ell(\beta) \approx \ell(\hat{\beta}) + \langle \nabla \ell(\hat{\beta}), \beta - \hat{\beta} \rangle + \frac{1}{2} (\beta - \hat{\beta})^T H\ell(\hat{\beta}) (\beta - \hat{\beta}).$$

Because $\nabla \ell(\hat{\beta}) = 0$, and because $\ell(\hat{\beta})$ is constant in β , we get

$$\begin{aligned} f_{\beta|Y=y}(\beta) &\propto \exp \left(\frac{1}{2} (\beta - \hat{\beta})^T H\ell(\hat{\beta}) (\beta - \hat{\beta}) \right) \\ &= \exp \left(-\frac{1}{2} (\beta - \hat{\beta})^T \left[-H\ell(\hat{\beta}) \right] (\beta - \hat{\beta}) \right). \end{aligned}$$

This is the unnormalized density of a multivariate normal distribution with mean $\hat{\beta}$ and covariance matrix $[-H\ell(\hat{\beta})]^{-1}$.

Let W be the $n \times n$ diagonal matrix with diagonal entries

$$W_{ii} = \frac{\exp(x_i^T \hat{\beta})}{(1 + \exp(x_i^T \hat{\beta}))^2}.$$

Then

$$-H\ell(\hat{\beta}) = X^T W X. \quad (3)$$

Therefore the posterior normal approximation is

$$\beta \mid Y = y \approx N\left(\hat{\beta}, (X^T W X)^{-1}\right). \quad (4)$$

The standard errors of $\hat{\beta}_0, \dots, \hat{\beta}_m$ are the square roots of the diagonal entries of $(X^T W X)^{-1}$.

It turns out that Bayesian inference based on this normal posterior approximation coincides with the usual large-sample frequentist inference for logistic regression. Thus the standard errors, confidence intervals, and z -tests reported by statistical software can also be viewed through this Bayesian normal approximation.

2.2.1 Frequentist Inference

For frequentist inference, one needs to understand the sampling distribution of the maximum likelihood estimator $\hat{\beta}$. Unlike ordinary linear regression, the logistic regression MLE does not have a closed-form expression. Consequently, its sampling distribution cannot be derived exactly in finite samples.

Instead, one relies on asymptotic theory for maximum likelihood estimators. Under suitable regularity conditions,

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, I(\beta)^{-1}),$$

where

$$I(\beta) = \lim_{n \rightarrow \infty} \frac{1}{n} X^T W X$$

is the Fisher information matrix. Equivalently,

$$\hat{\beta} \approx N(\beta, (X^T W X)^{-1}),$$

for large sample sizes. The asymptotic normality of maximum likelihood estimators is a general result (uses Taylor expansions as well); see van der Vaart [4]. Thus the standard errors reported by statistical software are obtained from the diagonal entries of $(X^T W X)^{-1}$, and confidence intervals and hypothesis tests are based on this asymptotic normal approximation.

Notice that this is a fundamentally frequentist justification and is asymptotic in nature. It is also exactly the same covariance matrix that appeared in the Bayesian normal approximation derived in the previous section. Thus Bayesian (with normal approximation) and frequentist inference lead to similar conclusion, atleast when the sample size is large.

2.3 Newton's Algorithm for Computing the MLE

The maximum likelihood estimator $\hat{\beta}$ is the value of β that maximizes the log-likelihood function $\ell(\beta)$. Unlike ordinary least squares, this maximizer cannot be written down in closed form. Instead, it must be computed numerically.

The most common approach is Newton's method. To understand the idea, suppose we wish to maximize a smooth function $f(x)$ of a single variable. Starting from a current guess x_0 , we approximate f near x_0 by its second-order Taylor expansion

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(x_0)(x - x_0)^2.$$

Thus, near x_0 , the function is approximated by a quadratic. Since a quadratic can be optimized explicitly, we maximize this approximation instead of the original function.

Differentiating the quadratic approximation and setting the derivative equal to zero gives

$$f'(x_0) + f''(x_0)(x - x_0) = 0.$$

Solving for x yields the update

$$x = x_0 - \frac{f'(x_0)}{f''(x_0)}.$$

This becomes the next iterate,

$$x_{r+1} = x_r - \frac{f'(x_r)}{f''(x_r)}.$$

In several dimensions exactly the same idea applies. At the current iterate $\beta^{(r)}$, we replace the log-likelihood by its quadratic Taylor approximation and maximize that quadratic.

More precisely, if β_0 is a point in parameter space, then the second-order Taylor approximation of $\ell(\beta)$ around β_0 is

$$\ell(\beta) \approx \ell(\beta_0) + \nabla \ell(\beta_0)^T (\beta - \beta_0) + \frac{1}{2}(\beta - \beta_0)^T H\ell(\beta_0)(\beta - \beta_0),$$

where $\nabla \ell(\beta_0)$ denotes the gradient vector and $H\ell(\beta_0)$ denotes the Hessian matrix.

To maximize this quadratic approximation, we differentiate with respect to β . Using the fact that

$$\nabla_{\beta} \{(\beta - \beta_0)^T H\ell(\beta_0)(\beta - \beta_0)\} = 2H\ell(\beta_0)(\beta - \beta_0),$$

the gradient of the quadratic approximation is

$$\nabla\ell(\beta_0) + H\ell(\beta_0)(\beta - \beta_0).$$

Setting this equal to zero yields

$$\nabla\ell(\beta_0) + H\ell(\beta_0)(\beta - \beta_0) = 0.$$

Solving for β gives

$$\beta = \beta_0 - (H\ell(\beta_0))^{-1} \nabla\ell(\beta_0).$$

This point is the maximizer of the quadratic approximation. Replacing β_0 by the current iterate $\beta^{(r)}$ produces the updates in Newton's method:

$$\beta^{(r+1)} = \beta^{(r)} - (H\ell(\beta^{(r)}))^{-1} \nabla\ell(\beta^{(r)}), \quad (5)$$

where

$$\nabla\ell(\beta) = \left(\frac{\partial\ell}{\partial\beta_0}, \dots, \frac{\partial\ell}{\partial\beta_m} \right)^T$$

is the gradient vector and

$$H\ell(\beta) = \left(\frac{\partial^2\ell}{\partial\beta_i\partial\beta_j} \right)_{i,j}$$

is the Hessian matrix.

Geometrically, Newton's method repeatedly replaces the log-likelihood by a quadratic approximation, computes the maximizer of that quadratic, and then moves to the resulting point. Near the true maximizer, the approximation is extremely accurate, and Newton's method converges very rapidly. Under suitable smoothness assumptions, Newton's method converges quadratically once the iterates are sufficiently close to the optimum; see Boyd and Vandenberghe [2]. This means that if

$$e_r = \|\beta^{(r)} - \beta^*\|$$

denotes the error at iteration r , then

$$e_{r+1} \leq C e_r^2$$

for some constant $C > 0$. Consequently, the number of correct digits roughly doubles at each iteration.

Now let us explicitly write out the Newton updates. As we saw in (2),

$$\nabla \ell(\beta) = \sum_{i=1}^n (y_i - \pi_i) x_i,$$

where

$$\pi_i = \frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)}.$$

Let π be the $n \times 1$ vector with entries $\pi_1, \dots, \pi_n$. These are the fitted probabilities at β .

Then

$$\nabla \ell(\beta) = X^T (y - \pi).$$

Also from (3),

$$H\ell(\beta) = -X^T W X,$$

where W is the diagonal matrix with entries

$$W_{ii} = \pi_i(1 - \pi_i).$$

Therefore Newton's update becomes

$$\beta^{(r+1)} = \beta^{(r)} + (X^T W X)^{-1} X^T (y - \pi).$$

This can be rewritten as

$$\beta^{(r+1)} = (X^T W X)^{-1} X^T W Z, \tag{6}$$

where

$$Z = X\beta^{(r)} + W^{-1}(y - \pi). \tag{7}$$

2.3.1 Iterative Reweighted Least Squares

The form of (6) should look familiar. Recall that in weighted least squares, given a response vector z , design matrix X , and positive definite weight matrix W , one minimizes

$$(z - X\beta)^T W (z - X\beta).$$

Differentiating with respect to β and setting the derivative equal to zero gives the normal equations

$$X^T W X \beta = X^T W z.$$

Provided $X^T W X$ is invertible, the solution is

$$\hat{\beta} = (X^T W X)^{-1} X^T W z.$$

Comparing this expression with (6), we see that the Newton update $\beta^{(r+1)}$ is exactly the weighted least squares estimator obtained by using

$$z = Z$$

as the response and

$$W = \text{diag}(\pi_1(1 - \pi_1), \dots, \pi_n(1 - \pi_n))$$

as the weight matrix.

Thus a single Newton step can be computed by solving a weighted least squares problem. After computing a new estimate $\beta^{(r)}$, we must recompute the fitted probabilities π_i , construct a new weight matrix W , construct a new pseudo-response vector Z , and solve another weighted least squares problem.

In other words, logistic regression is solved by repeatedly fitting weighted least squares models, with the weights changing from one iteration to the next. For this reason the algorithm is known as *iteratively reweighted least squares* (IRLS).

The weights have a natural interpretation. Since

$$W_{ii} = \pi_i(1 - \pi_i),$$

observations whose fitted probabilities are near 1/2 receive the largest weights, while observations whose fitted probabilities are very close to 0 or 1 receive relatively small weights. The algorithm therefore pays the most attention to observations near the current classification boundary, where the uncertainty is greatest.

3 Generalized Linear Models

Linear regression and logistic regression both belong to a broader family known as *generalized linear models* (GLMs). The GLM framework provides a unified way to model a wide variety of response variables and is one of the classical ideas in statistics [3].

In linear regression, we model the conditional mean directly:

$$E(Y_i \mid X_i = x_i) = x_i^T \beta. \tag{8}$$

For a binary response, with $p_i = E(Y_i | X_i = x_i)$, instead of modeling the probability directly, logistic regression models the log-odds:

$$\log\left(\frac{p_i}{1-p_i}\right) = x_i^T \beta. \quad (9)$$

This transformation automatically guarantees that the fitted probabilities remain between 0 and 1.

This idea of using some transformation of the mean response and then modelling it as a linear function of X is useful more generally and gives rise to GLM. A generalized linear model consists of two ingredients:

1. A probability distribution for the response variable, like Gaussian, Bernoulli, Poisson etc.
2. A link function applied to the conditional mean of the response which is then modelled as a linear function of X .

If $\mu_i = E(Y_i | X_i = x_i)$ then the general GLM form is

$$g(\mu_i) = x_i^T \beta, \quad (10)$$

where g is called the *link function*.

3.1 Poisson Regression

Many response variables represent counts rather than continuous measurements. Examples include the number of bicycles rented in an hour, the number of insurance claims filed by a customer, the number of visits to a website, and the number of hospital admissions in a day. Such data are often modeled using the Poisson distribution.

A random variable Y is said to have a Poisson distribution with parameter $\lambda > 0$ if

$$P(Y = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots \quad (11)$$

The parameter λ is both the mean and the variance,

$$E(Y) = \lambda, \quad \text{Var}(Y) = \lambda. \quad (12)$$

As a concrete example, suppose Y_i denotes the number of bicycles rented during a particular hour. We might expect the average number of rentals to depend on variables such

as temperature, season, weather conditions, and the time of day. Since the response consists of nonnegative counts, ordinary linear regression is not ideal because it can produce negative fitted values. Also, it is possible that the variance of Y is not the same for all X (heteroscedastic), if we think that the variances are roughly equal to the mean, along with Y being nonnegative integer valued, then modelling Y as a Poisson random variable becomes natural.

Poisson regression assumes that

$$Y_i \mid X_i = x_i \sim \text{Poisson}(\lambda_i). \quad (13)$$

Since λ_i must be positive, we model its logarithm as a linear function of the predictors:

$$\log(\lambda_i) = x_i^T \beta. \quad (14)$$

Equivalently,

$$\lambda_i = \exp(x_i^T \beta). \quad (15)$$

This guarantees that the fitted mean is always positive.

3.2 Interpreting the Coefficients

Suppose predictor x_j increases by one unit while all other predictors remain fixed.

Then

$$\log(\lambda) \longrightarrow \log(\lambda) + \beta_j, \quad (16)$$

which implies

$$\lambda \longrightarrow e^{\beta_j} \lambda. \quad (17)$$

Thus e^{β_j} represents the multiplicative change in the expected count associated with a one-unit increase in predictor j . Again we really need to interpret this as the multiplicative change adjusted for all the other predictors. This multiplicative interpretation is often more natural for count data than the additive interpretation encountered in ordinary linear regression.

3.3 Maximum Likelihood Estimation

For independent observations, the likelihood is

$$L(\beta) = \prod_{i=1}^n e^{-\lambda_i} \frac{\lambda_i^{y_i}}{y_i!}, \quad \lambda_i = \exp(x_i^T \beta). \quad (18)$$

As with logistic regression, the maximum likelihood estimator cannot be written in closed form and is computed numerically using Newton's method.

3.4 Other Examples of GLMs

Response Type	Distribution	Typical Mean Constraint	Common Link
Continuous measurements	Gaussian	$-\infty < \mu < \infty$	Identity
Binary outcomes	Bernoulli	$0 < p < 1$	Logit
Count data	Poisson	$\lambda > 0$	Log
Positive continuous responses	Gamma	$\mu > 0$	Log
Overdispersed counts	Negative Binomial	$\mu > 0$	Log

Gamma regression is frequently used for positive continuous quantities such as insurance costs, waiting times, and rainfall amounts. Negative binomial regression is commonly used for count data when the variance substantially exceeds the mean, a phenomenon known as *overdispersion*.

3.5 Summary

Linear regression models a conditional mean, logistic regression models a probability, and Poisson regression models a count rate. Generalized linear models reveal that all three are examples of the same underlying idea:

Choose an appropriate probability model for the response, transform its mean using a suitable link function, and then model the transformed mean as a linear function of the predictors.

Remark 3.1. *The examples discussed in this chapter are instances of a much broader theory based on exponential-family distributions. One of the remarkable features of this theory is that many seemingly different statistical models can be studied using the same mathematical tools. In particular, maximum likelihood estimation, Newton's method, iteratively reweighted least squares (IRLS), standard error calculations, and asymptotic normality can all be developed simultaneously for a large class of models. From this viewpoint, linear regression, logistic regression, and Poisson regression are not three different techniques*

but three instances of the same modelling principle. The choice of response distribution and link function determines the model, while estimation and inference are carried out using essentially the same ideas. This perspective was one of the major achievements of twentieth-century statistics and is one of the reasons generalized linear models remain so widely used today; see [1] for a modern introduction to GLM.

References

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